

By the defn. of existential quantifier, $\exists x P(x)$ is False. By the defn. of conditional, $A \rightarrow \exists x P(x)$ is False.

$$\boxed{\therefore \exists x (A \rightarrow P(x)) \text{ is False} \Rightarrow A \rightarrow \exists x P(x) \text{ is False}}$$

Let, $A \rightarrow \exists x P(x)$ be False. By the defn. of conditional, A is True and $\exists x P(x)$ is False. By the defn. of negation, $\neg \exists x P(x)$ is True. By De Morgan's law of quantifiers, $\forall x (\neg P(x))$ is True. By the defn. of universal quantifier, for all elements a in the domain, $\neg P(a)$ is True. By the defn. of negation, $\neg (\neg P(a))$ is False. By double negation law, $P(a)$ is False for every element a in the domain. By defn. of conditional, $A \rightarrow P(a)$ is False, for every element a in the domain. By defn. of existential quantifier, $\exists x (A \rightarrow P(x))$ is False.

$$\boxed{\therefore A \rightarrow \exists x P(x) \text{ is False} \Rightarrow \exists x (A \rightarrow P(x)) \text{ is False}}$$

$$\boxed{\therefore \text{They are logically equivalent } \square}$$

5) Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is non-empty.

a) $\forall x (P(x) \rightarrow A) \equiv \exists x P(x) \rightarrow A$.

Let, $\forall x (P(x) \rightarrow A)$ be False. By defn. of universal quantifier, there exists an element a in the domain such that $P(a) \rightarrow A$ is False. By defn. of conditional, $P(a)$ is True and A is False. By defn. of existential quantifier, $\exists x P(x)$ is True. By defn. of conditional, $\exists x P(x) \rightarrow A$ is False.

$$\boxed{\therefore \forall x (P(x) \rightarrow A) \text{ is False} \Rightarrow \exists x P(x) \rightarrow A \text{ is False}}$$

Let, $\exists x P(x) \rightarrow A$ be False. By defn. of conditional, $\exists x P(x)$ is True and A is False. By defn. of existential quantifier, there exists at least one element a in the domain such that $P(a)$ is True. By the